

Homework 2

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Question 1

Consider as shortest path problem with four states $\mathcal{S} = \{1, 2, 3, 4\}$ and two actions per state $\mathcal{A} = \{1, 2\}$. Let state 4 be a cost-free terminal state. Otherwise, the cost per transition depends only on action, it is

$$g(a) = \begin{cases} -1 & \text{if } a = 1, \\ -2 & \text{if } a = 2. \end{cases}$$

The transition function for $a = 1$ is $p_{ii}(1) = 0.9$ and $p_{i,i+1}(1) = 0.1$ for $i = 1, 2, 3$ and for $a = 2$ it is $p_{11}(2) = p_{12}(2) = 0.5$ and $p_{i,i+1}(2) = p_{i,i-1}(2) = 0.5$ for $i = 2, 3$.

- a) Compute the value of the stationary policies $\mu_1(s) = 1$ and $\mu_2(s) = 2$.
- b) Find the optimal policy.
- c) Let T be the optimal Bellman operator for the problem. Find a norm $\|\cdot\|$ and a parameter $\beta \in [0, 1)$ such that T is β -contractive in $\|\cdot\|$.

Question 2

Consider a Markov reward process with states $\mathcal{S} = \{0, 1, \dots, S\}$, transition matrix P and reward function $R : \mathcal{S} \rightarrow \mathbb{R}$. Here the state 0 is an absorbing state, i.e., $R(0) = 0$ and $P(0, 0) = 1$ and $P^k(i, 0) > 0$ for all $i \in \mathcal{S}$ and some $k \in \mathbb{N}$. Consider the following value function for the shortest path problem:

$$v(s) = \mathbb{E} \left[\sum_{k=0}^{\infty} R_k \mid S_0 = s \right].$$

We sometimes write v and r as column vectors $v, r \in \mathbb{R}^{S+1}$ where

$$v = (v(0), v(1), \dots, v(S)) \quad \text{and} \quad r = (r(0), r(1), \dots, r(S))$$

and sometimes we would like to exclude the value/reward at the absorbing state, then we write $\hat{v} : \mathcal{S}_+ \rightarrow \mathbb{R}^S$ and $\hat{r} : \mathcal{S}_+ \rightarrow \mathbb{R}^S$, where $\mathcal{S}_+ = \mathcal{S} \setminus \{0\}$.

We now go through the steps of computing the value $v(s)$ by solving a linear system of equations.

- a) Show that the transition matrix P has the following structure

$$P = \begin{bmatrix} 1 & \mathbf{0} \\ P_1 & P_2 \end{bmatrix}$$

where $P_1 \in \mathbb{R}^{S \times 1}$, $P_2 \in \mathbb{R}^{S \times S}$, and $\mathbf{0} = [0, \dots, 0] \in \mathbb{R}^{1 \times S}$.

- b) Prove that $I - P_2$ is nonsingular, where $I \in \mathbb{R}^{S \times S}$ is the identity matrix.
c) Prove that the value is

$$v = \begin{bmatrix} v(0) \\ \hat{v} \end{bmatrix} = \begin{bmatrix} 0 \\ (I - P_2)^{-1} \hat{r} \end{bmatrix}.$$

Question 3

Consider the same setting as in Question 2. Another approach for computing the value is the value iteration algorithm. The algorithm progresses as

$$v^{k+1} = r + P v^k$$

where k is the iteration index, v^k is the estimate of the value at iteration k , where the initialization is any $v^0 \in \mathbb{R}^{S+1}$ with $v^0(0) = 0$ (the value of the state 0 is always 0). The goal of this problem is to prove that the value evaluation algorithm is contractive and converges at a linear rate.

- a) Use the sparsity structure of P (see Exercise 2.a) to show that we can write the algorithm equivalently as

$$\hat{v}^{k+1} = \hat{r} + P_2 \hat{v}^k$$

where $\hat{v}^0 \in \mathbb{R}^S$ is some initialization.

- b) Define $N : \mathcal{S}_+ \rightarrow \mathbb{R}$ as the expected number of times state s is visited before reaching the absorbing state 0. Show that following holds for all $s \in \mathcal{S}_+$

$$N(s) = 1 + \sum_{s' \in \mathcal{S}_+} P(s, s') N(s').$$

Hint: Show how we can create a new MRP whose value is N , thus we can compute N by solving a linear system similarly as in Exercise 2.

- c) Consider the weighted norm

$$\|w\|_N = \max_{s \in \mathcal{S}_+} \frac{|w(s)|}{N(s)}$$

on $\mathbb{R}^S$. Show that the function

$$B(w) = \hat{r} + P w$$

is a α -contraction where

$$\alpha = \max_{s \in \mathcal{S}_+} \frac{N(s) - 1}{N(s)}.$$

In particular, show that $\alpha \in [0, 1)$ and that

$$\|B(w_1) - B(w_2)\|_N \leq \alpha \|w_1 - w_2\|_N$$

for all $w_1, w_2 \in \mathbb{R}^S$. Conclude that the value estimation algorithm converges at the linear rate

$$\|v^k - v\|_N \leq \alpha^k \|v^0 - v\|_N$$

- d) Use the above to show that we find an ϵ -accurate solution (i.e., such that $\|v^k - v\|_N \leq \epsilon$ after

$$k = N_{\max} \log \left(\frac{\|v^0 - v\|_N}{\epsilon} \right) \text{ iterations}$$

where $N_{\max} := \max_{s \in \mathcal{S}_+} N(s)$.

Question 4

Consider a Markov reward process with states $\mathcal{S} = \{1, \dots, S\}$, transition matrix P and reward function $R : \mathcal{S} \rightarrow \mathbb{R}$. We wish to compute the discounted value function

$$v(s) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k R_k \mid S_0 = s \right].$$

As we have seen in the book and the lectures, for a given P and R we have $v = (I - \gamma P)^{-1} R$. However, sometimes we might only have an approximation of P and R . In this problem we derive some bounds on the estimated value when it is computed based on such an approximate model.

- a) Suppose that we know the true transition matrix P , but only have an approximation $\hat{R}$ of the reward function. Suppose that $\|R - \hat{R}\|_{\infty} \leq \epsilon$ and we compute the approximate value

$$\hat{v} = (I - \gamma P)^{-1} \hat{R}.$$

Prove the following bound on the approximation accuracy

$$\|\hat{v} - v\|_{\infty} \leq \frac{\epsilon}{1 - \gamma}.$$

Note: For a vector norm $\|\cdot\|$ on $\mathbb{R}^d$ we can define an induced matrix norm on $\mathbb{R}^{d \times d}$ as follows:

$$\|A\| = \sup_{x \in \mathbb{R}^d} \frac{\|Ax\|}{\|x\|}$$

for $\mathbb{R}^{d \times d}$. In the case of the ∞ -norm, the induced matrix norm is

$$\|A\|_{\infty} = \max_{i=1,2,\dots,d} \sum_{j=1}^d |A_{ij}|.$$

For induced matrix norms we have $\|Ax\| \leq \|A\| \|x\|$.

Hint: It is possible to use the following identity

$$(I + B)^{-1} + B(I + B)^{-1} = (I + B)(I + B)^{-1} = I$$

which holds true when $(I + B)^{-1}$ exists, to prove that

$$\|(I + B)^{-1}\| \leq \frac{1}{1 - \|B\|},$$

provided that $\|B\| < 1$ and $\|B\|$ is induced matrix norm.

- b) Suppose that we have an approximate $\hat{v}$ of the true value function. Suppose that $\|\hat{v} - B(\hat{v})\|_{\infty} \leq \epsilon$ where $B(v) = R + \gamma P v$ is the Bellman operator. Prove that

$$\|\hat{v} - v\|_{\infty} \leq \frac{\epsilon}{1 - \gamma}.$$

Note: This bound can be useful because if we have an estimator then we can always calculate $\|v - B(v)\|_{\infty}$ even if we don't know the true value v .

- c) Suppose that we know the true reward R , but we only have an approximation of $\hat{P}$ of the transition matrix. Suppose that $\|P - \hat{P}\|_{\infty} \leq \epsilon$ and we compute the approximate value

$$\hat{v} = (I - \gamma \hat{P})^{-1} R.$$

Prove the following bound on the approximation accuracy

$$\|\hat{v} - v\|_{\infty} \leq \frac{\gamma \epsilon}{1 - \gamma}.$$

Question 5

For both discounted and shortest-path MRP, under what conditions is the dynamic programming algorithm more efficient than solving the linear system and vice versa. When are the two algorithms comparable?

Question 6

Consider the optimal Bellmann operator for a discounted Markov Decision Process

$$\mathcal{T}(Q)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left(\max_{a' \in \mathcal{A}} Q(s', a') \right),$$

where $\gamma \in [0, 1)$. Prove that $\mathcal{T}(\cdot)$ is contractive in the ∞ -norm with parameter γ and fixed point Q^* , where Q^* is the optimal Q-table. Use these results to establish a convergence rate on the value iteration for discounted MRP.

Question 7

Consider the Markov Reward Process with state space $\mathcal{S} = \{1, 2\}$ and reward vector and transition matrix

$$r = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

We now compute the gain and bias using three different approaches.

- a) Compute the gain and bias of the MRP using the definition. Recall that the gain is defined as

$$g(s) = \lim_{K \rightarrow \infty} \frac{1}{K} \mathbf{E} \left[\sum_{k=1}^K r(S_k) \middle| S_1 = s \right]$$

and the bias as

$$h(s) = \text{C-lim}_{k \rightarrow \infty} \mathbf{E} \left[\sum_{k=1}^{\infty} r(S_k) - g(S_k) \middle| S_1 = s \right].$$

- b) Compute the stationary distribution of the Markov chain and use it to compute the gain.
c) Compute the discounted rewards

$$v_{\lambda}(s) = \mathbf{E} \left[\sum_{k=0}^{\infty} \lambda^k r(S_k) \middle| S_0 = s \right]$$

for $\lambda \in [0, 1)$. Use the values $v_{\lambda}(s)$ to compute the gain.

- d) Consider the linear system

$$g\mathbf{1} + (I - P)h = r$$

or

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} g \\ h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

where $\mathbf{1} = [1, 1]^T$ and I is the 2×2 identity matrix. Prove that all solutions of this system have the same value of g and compute that value of g .

Hint: look at the null space.

Question 8

Consider a average reward Markov reward process $(\mathcal{S}, P, r)$ with S . Recall the span semi-norm on $\mathbb{R}^S$:

$$\mathbf{sp}(v) = \max_{s \in \mathcal{S}} v(s) - \min_{s \in \mathcal{S}} v(s).$$

Assume that P has the property that there exists $\gamma \in [0, 1)$ such that

$$\mathbf{sp}(Pv) \leq \gamma \mathbf{sp}(v)$$

for all v . Under this assumption, we prove that the Value Iteration (VI) converges to the gain g^* and a bias h^* satisfying

$$g^* \mathbf{1} + h^* = r + Ph^*.$$

We do it in the following 3 steps.

- a) Consider the VI where we start with some initial bias estimate h^0 and after that, for each iteration k , we perform the update

$$h^{k+1} = L(h^k) := r + Ph^k.$$

Prove that that

$$\mathbf{sp}(h^k - h^*) \leq \gamma^k \mathbf{sp}(h^0 - h^*).$$

- b) Prove that if $\mathbf{sp}(h - h^*) \leq \epsilon$ then there exists a scalar c and a vector e where $\|e\|_\infty \leq \epsilon/2$ where $h = h^* + c\mathbf{1} + e$.
- c) Define $g^k = L(h^k) - h^k$, prove that

$$\|g^k - g^* \mathbf{1}\|_\infty \leq 2\mathbf{sp}(h^0 - h^*)\gamma^k.$$